School of Future Tech

Case Study Report

on

Case Study 143: Smart Waste Collection & Recycling Management System

by

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Case Study: Smart Waste Collection & Recycling Management System

Github Link: https://github.com/Ankitraj-17/DSA-Final

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1. Introduction to the Case Study

As modern municipalities undergo rapid urbanization, the volume of solid waste generated daily escalates exponentially. Municipal corporations are tasked with coordinating collection routines under constraints of restricted budgets, limited fleets, fuel cost fluctuations, and environmental goals.

Traditional municipal solid waste management systems suffer from high operational overhead due to fixed schedules and static routing. Collection vehicles visit zones without knowing bin fill status, leading to empty trips or overflows. Furthermore, citizens complain of long response times for service requests, and managing databases of thousands of bins becomes slow if search indexes degrade.

This case study designs and implements a **Smart Waste Collection & Recycling Management System** in C++ using custom-designed Data Structures and Algorithms. The platform integrates IoT sensor simulation, shortest-path road navigation, dynamic priority queues, and self-balancing tree databases to provide a highly scalable, real-time command dashboard.

Efficient algorithmic waste management is applicable in:

Smart Cities: Municipal route optimization and sensor-driven garbage collection.
Logistics & Fleet Management: Real-time asset checks, license lookups, and scheduling.
Environmental Sustainability: Dynamic estimation of recyclable volumes and resource deployment.

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2. Problem Statement (Abstract)

Managing urban waste requires scheduling vehicles to clean bins across different districts (commercial, industrial, residential) and handling citizen service tickets. The primary technical challenges include:

1. **Inefficient Scheduling:** Sorting collection orders randomly or by static location causes bins with high waste levels to overflow, generating odors and hazards. Bins must be sorted dynamically by priority scores based on fill percentage and capacity.
2. **Logarithmic Retrieval Bounds:** Inserting sequential bin IDs into standard binary search trees causes skewing, degrading lookup times from O(log N) to O(N) linear scans. Database structures must maintain balance factors.
3. **Logistics Route Optimization:** Manually routing collection vehicles leads to excessive fuel consumption. A graph representation of city intersections must be queried to calculate the shortest path.
4. **Log Retrieval & Undo Actions:** Operational mistakes (such as duplicate entries or incorrect routing) require a recovery mechanism that lists and rolls back changes in LIFO order.
5. **Fair Ticket Handling:** Citizen complaints must be processed in a strict FIFO manner to prevent long wait times.

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3. Case Study Design

The smart waste platform is designed as an interactive command-line dashboard that operates entirely on custom-built pointer structures, replacing STL dependencies to demonstrate low-level DSA logic.

Input

Waste Bin Array: Dynamic entries representing bins with:
- ID (integer key)
- Location (string name)
- Fill Level (0 to 100%)
- Capacity (in liters)
- Waste Type (Organic, General, Recyclable, Hazardous)

City Intersections Map: A weighted, undirected graph network consisting of 10 vertices (Central Depot, Recycling Center, and key zones) and 12 weighted edges (distances in km).
Fleet Vehicles: License plate strings, driver names, truck type, and status flags.
Service Tickets: Citizen name, location, problem description, and urgency ratings.

Constraints

Memory Bounds: Explicit deletion of dynamically allocated tree nodes, queue items, and stack items to prevent leaks.
Sequential Sorting Constraint: Prioritization must run sorting operations in-place without standard templates (`std::sort`).
Balanced Trees Constraint: Tree height must remain bounded to O(log N) under sequential insertions.

Real-World Mapping

Ultrasonic Sensors: Send fill levels dynamically to update bin records.

License Plates: Act as unique keys for polynomial hashing to track active trucks.
Graph Vertices: Depot, zones, and transfer stations.

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4. Methods & Algorithms Technology Applied

Feature	Data Structure / Algorithm	Justification
Collection Scheduling	**Sorting Algorithms (Merge/Quick)**	Orders the collection zones by waste generation level to prioritize high-need areas efficiently.
Record Retrieval	**Searching Algorithms (Binary Search)**	Allows quick, logarithmic-time lookup of historical recycling and collection records from sorted databases.
Recovery Procedures	**Stack**	Uses LIFO (Last-In-First-Out) ordering. The most recent operational change is at the top, perfectly mimicking the behavior of an 'Undo' button.
Service Requests	**Queue**	Enforces FIFO (First-In-First-Out) processing, guaranteeing that citizen or municipal requests are processed strictly in the order they were submitted.
Asset Identification	**Hash Table**	Provides near-instant (O(1)) lookup times to verify if a given physical asset (RFID or GPS ID) already exists or to find its status.
Waste Database	**BST / AVL Trees**	Automatically balances the collection database so that insertion, deletion, and searching remain fast and predictable at O(log N).
Route Optimization	**Graph & Dijkstra's Algorithm**	Models city intersections as vertices and roads as edges with distance weights. Dijkstra’s guarantees the shortest and most cost-effective path for garbage trucks.

Time & Space Complexity

Component	Operation	Time Complexity	Space Complexity
Sorting Algorithms	Sort Schedules	O(N log N)	O(N) or O(log N)

Component	Operation	Time Complexity	Space Complexity
Searching Algorithms	Record Lookup	O(log N)	O(1)
Stack	Push / Pop	O(1)	O(A) where A is number of stored actions
Queue	Enqueue / Dequeue	O(1)	O(S) where S is requests in queue
Hash Table	Insert / Lookup	O(1) average, O(C) worst	O(C) where C is unique assets
BST/AVL Tree	Insert / Search	O(log D)	O(D) where D is collection points
Graph	Add Edge	O(1) average	O(V + E) where V=intersections, E=roads
Dijkstra / BFS	Route finding	O((V + E) log V)	O(V) for priority queue and visited sets

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5. Problem Statement / Case Study Implementation Details and Snapshots

The core logic of the system is written in `smart_waste_system.cpp`. The platform avoids pre-built standard templates, creating custom objects for the data layers.

Core Structure Implementations

1. Custom Singly-Linked Stack (LIFO Recovery Log)

The recovery mechanism logs all operational modifications using dynamic singly linked nodes:

2. Custom FIFO Queue (Service Request Manager)

Citizen tickets are processed in order using a linked queue structure tracking head and tail nodes:

3. Custom AVL Self-Balancing Tree (Balanced Database Storage)

Primary database storage inherits standard binary search operations but wraps them in rotations to maintain logarithmic height:

```
None
struct TreeNode {
```

```
    WasteBin bin;
    TreeNode *left, *right;
    int height;
};

class AVL {
public:
    TreeNode* root = nullptr;
    int count = 0;
    int getH(TreeNode* n) { return n ? n->height : 0; }
    int getBF(TreeNode* n) { return n ? getH(n->left) -
getH(n->right) : 0; }
    void updH(TreeNode* n) { if (n) n->height = 1 +
maxVal(getH(n->left), getH(n->right)); }

    TreeNode* rotR(TreeNode* y) {
        TreeNode* x = y->left, *T2 = x->right;
        x->right = y;
        y->left = T2;
        updH(y);
        updH(x);
        return x;
    }
    TreeNode* rotL(TreeNode* x) {
        TreeNode* y = x->right, *T2 = y->left;
        y->left = x;
        x->right = T2;
        updH(x);
        updH(y);
        return y;
    }
    TreeNode* bal(TreeNode* n) {
        updH(n);
        int bf = getBF(n);
        if (bf > 1) {
            if (getBF(n->left) < 0) n->left = rotL(n->left);
            return rotR(n);
        }
        if (bf < -1) {
```

```
            if (getBF(n->right) > 0) n->right = rotR(n->right);
            return rotL(n);
        }
        return n;
    }
    TreeNode* ins(TreeNode* n, WasteBin b) {
        if (!n) { count++; return new TreeNode{b, nullptr,
nullptr, 1}; }
        if (b.id < n->bin.id) n->left = ins(n->left, b);
        else if (b.id > n->bin.id) n->right = ins(n->right, b);
        else return n;
        return bal(n);
    }
    void insert(WasteBin b) { root = ins(root, b); }
};
```

4. Custom Hash Table with Collision Chaining (License lookup)

Fleet assets are indexed in a size-53 bucket structure using a polynomial rolling hash:

5. Shortest Route Optimization (Dijkstra's Pathfinder)

Computes shortest paths on the city graph using relaxation over arrays:

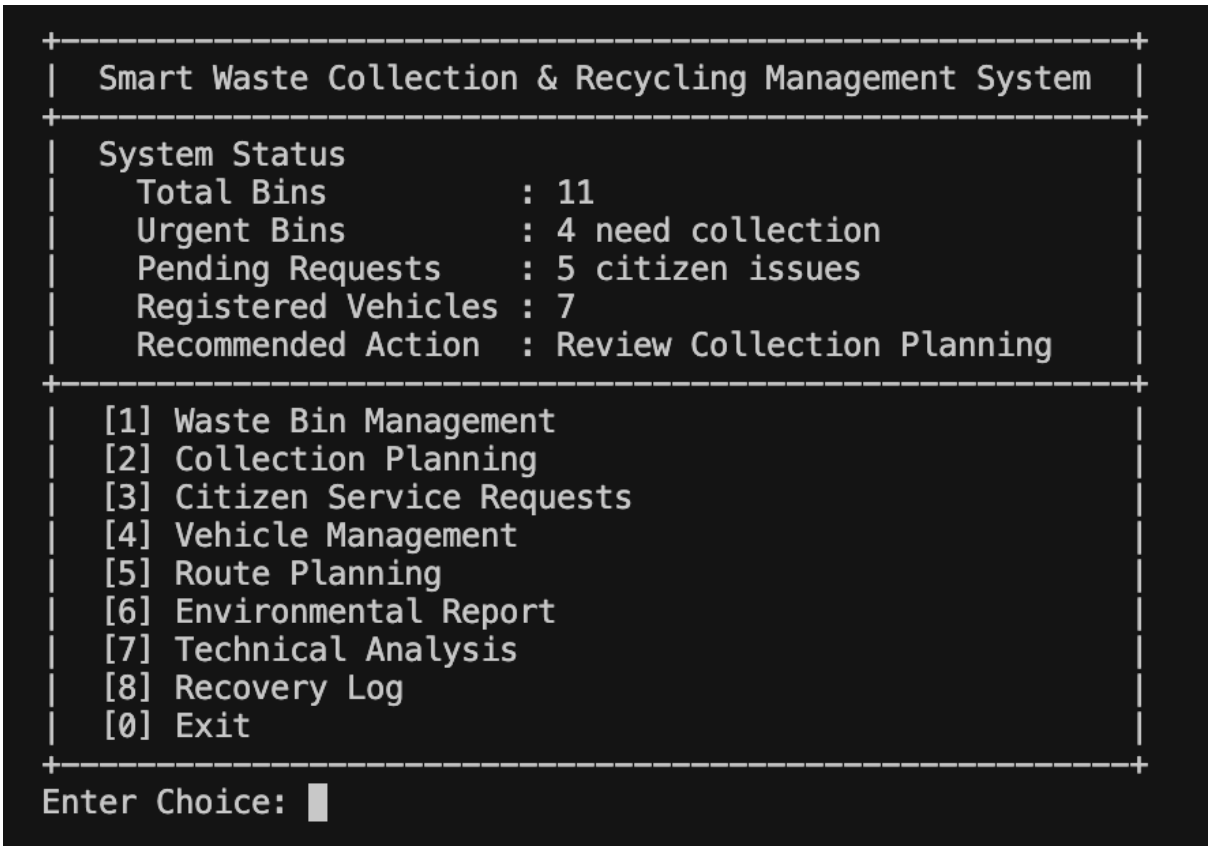
None

```
void dijkstra(int src, int dst) {
    int dist[maxVertices], prev[maxVertices];
    bool visited[maxVertices];
    for (int i = 0; i < V; i++) {
        dist[i] = 999999;
        prev[i] = -1;
        visited[i] = false;
    }
    dist[src] = 0;
    for (int count = 0; count < V - 1; count++) {
        int minDist = 999999, u = -1;
        for (int v = 0; v < V; v++) {
            if (!visited[v] && dist[v] < minDist) {
                minDist = dist[v];
                u = v;
            }
        }
```

```
        if (u == -1) break;
        visited[u] = true;
        for (size_t i = 0; i < adjList[u].size(); i++) {
            int neighbor = adjList[u][i].dest;
            int weight = adjList[u][i].weight;
            if (!visited[neighbor] && dist[u] + weight <
dist[neighbor]) {
                dist[neighbor] = dist[u] + weight;
                prev[neighbor] = u;
            }
        }
    }
}
```

5.1. SnapShorts

1. Main Dashboard



2. Waste Bin Management

```
+-------------------------------------------------+
|               Waste Bin Management              |
+-------------------------------------------------+
|  [1] View All Bins                              |
|  [2] Add New Bin                                |
|  [3] Search Bin by ID                           |
|  [4] Search Bin by Location                     |
|  [5] Remove Bin                                 |
|  [0] Back                                       |
+-------------------------------------------------+
Enter choice: 1

=== Waste Bin Records ===
+--------+----------------------+--------+----------+-------------+-----------+
| ID     | Location             | Fill%  | Capacity | Type        | Status    |
+--------+----------------------+--------+----------+-------------+-----------+
| 2      | 101                  | 2%     | 3L       | 6           | Normal    |
| 101    | Main Street          | 92%    | 500L     | Organic     | URGENT    |
| 103    | Industrial Area      | 88%    | 1000L    | General     | URGENT    |
| 104    | Residential North    | 30%    | 400L     | Organic     | Normal    |
| 105    | Residential South    | 76%    | 400L     | Recyclable  | Moderate  |
| 106    | Market District      | 95%    | 600L     | General     | URGENT    |
| 107    | Commercial Hub       | 55%    | 750L     | Hazardous   | Moderate  |
| 108    | University Area      | 40%    | 350L     | Recyclable  | Normal    |
| 109    | Town Square          | 83%    | 450L     | Organic     | URGENT    |
| 110    | Riverside Zone       | 62%    | 550L     | General     | Moderate  |
| 1003   | mumbai               | 50%    | 7L       | wet         | Moderate  |
+--------+----------------------+--------+----------+-------------+-----------+
Total Bins: 11 | Primary Storage Depth: 4

Press Enter to continue...
```

3. Collection Planning

```
+-------------------------------------------------+
|                Vehicle Management               |
+-------------------------------------------------+
|  [1] View Registered Vehicles                   |
|  [2] Search Vehicle                             |
|  [3] Add New Vehicle                            |
|  [0] Back                                       |
+-------------------------------------------------+
Enter choice: 1

=== Registered Vehicles ===
+----------------+--------------------+--------------------+--------------+----------+
| Plate Number   | Driver Name        | Vehicle Type       | Status       | Zone     |
+----------------+--------------------+--------------------+--------------+----------+
| 111            | 2                  | 1                  | 2            | 1        |
| GV-TRK-001     | Rajesh Kumar       | Compactor          | Available    | Central  |
| GV-TRK-002     | Suresh Mehta       | Tipper             | On Route     | North    |
| GV-TRK-003     | Amit Sharma        | Compactor          | Maintenance  | South    |
| GV-RCY-001     | Priya Singh        | Recycling Van      | Available    | East     |
| GV-RCY-002     | Deepak Verma       | Recycling Van      | On Route     | West     |
| GV-TRK-009     | Karan Singh        | Compactor          | Available    | North    |
+----------------+--------------------+--------------------+--------------+----------+

Press Enter to continue...
```

4. Route Planning

```
Enter choice: 3

+--------------------------------------------------+
|                  Route Planning                  |
+--------------------------------------------------+
|  [1] Show City Map                               |
|  [2] View Reachable Service Areas                |
|  [3] Check City Network Coverage                 |
|  [4] Find Shortest Collection Route              |
|  [0] Back                                        |
+--------------------------------------------------+
Enter choice: 1

=== City Connections Map ===
+--------------------+-----------------------------------------------------+
| Source Area        | Connected Areas (Distance)                          |
+--------------------+-----------------------------------------------------+
| Central Depot      | Main Street (2km), Park Avenue (3km)                |
| Main Street        | Central Depot (2km), Industrial Area (4km), Reside  |
| Park Avenue        | Central Depot (3km), Residential South (3km), Mark  |
| Industrial Area    | Main Street (4km), Commercial Hub (1km)             |
| Residential North  | Main Street (2km), Commercial Hub (3km)             |
| Residential South  | Park Avenue (3km), University Area (2km)            |
| Market District    | Park Avenue (5km), Recycling Center (4km)           |
| Commercial Hub     | Industrial Area (1km), Residential North (3km), Un  |
| University Area    | Residential South (2km), Commercial Hub (6km), Rec  |
| Recycling Center   | Market District (4km), University Area (3km)        |
+--------------------+-----------------------------------------------------+

Press Enter to continue...
```

6. Environment Report

```
=== Environmental Report ===
+----------------------------------------+------------------+
| Metric                                 | Value            |
+----------------------------------------+------------------+
| Total Bins Monitored                   | 11               |
| Bins Needing Collection (>=80%)        | 4                |
| Estimated Waste Volume                 | 3609 L           |
| Recyclable Waste Share                 | 12.3%            |
| Recycling Share Progress               | [=               |
| Pending Citizen Issues                 | 5                |
| Registered Vehicles                    | 7                |
+----------------------------------------+------------------+

Press Enter to continue...
```

7. Technical Analysis

```
+-------------------------------------------------+
|               Technical Analysis                |
+-------------------------------------------------+
|  [1] Primary Storage Status                     |
|  [2] Backup Storage Status                      |
|  [3] Storage Comparison                         |
|  [4] Feature Performance Summary                |
|  [0] Back                                       |
+-------------------------------------------------+
Enter choice: 1

=== Primary Storage Status (AVL Tree) ===
+--------+-----------------------+--------+------------+-------------+-----------+
| ID     | Location              | Fill%  | Capacity   | Type        | Status    |
+--------+-----------------------+--------+------------+-------------+-----------+
| 2      | 101                   | 2%     | 3L         | 6           | Normal    |
| 101    | Main Street           | 92%    | 500L       | Organic     | URGENT    |
| 103    | Industrial Area       | 88%    | 1000L      | General     | URGENT    |
| 104    | Residential North     | 30%    | 400L       | Organic     | Normal    |
| 105    | Residential South     | 76%    | 400L       | Recyclable  | Moderate  |
| 106    | Market District       | 95%    | 600L       | General     | URGENT    |
| 107    | Commercial Hub        | 55%    | 750L       | Hazardous   | Moderate  |
| 108    | University Area       | 40%    | 350L       | Recyclable  | Normal    |
| 109    | Town Square           | 83%    | 450L       | Organic     | URGENT    |
| 110    | Riverside Zone        | 62%    | 550L       | General     | Moderate  |
| 1003   | mumbai                | 50%    | 7L         | wet         | Moderate  |
+--------+-----------------------+--------+------------+-------------+-----------+

Preorder Traversal: 104 101 2 103 108 106 105 107 110 109 1003

=== AVL Node Connections Table ===
+------------+-----------------+-----------------+
| Node ID    | Left Child      | Right Child     |
+------------+-----------------+-----------------+
| 104        | Bin 101         | Bin 108         |
| 101        | Bin 2           | Bin 103         |
| 2          | None            | None            |
| 103        | None            | None            |
| 108        | Bin 106         | Bin 110         |
| 106        | Bin 105         | Bin 107         |
| 105        | None            | None            |
| 107        | None            | None            |
| 110        | Bin 109         | Bin 1003        |
| 109        | None            | None            |
| 1003       | None            | None            |
+------------+-----------------+-----------------+

Press Enter to continue...
```

8. Recovery Log

```
+-------------------------------------------------+
|                  Recovery Log                   |
+-------------------------------------------------+
|  [1] View Log History                           |
|  [2] Clear Last Log Entry                       |
|  [0] Back                                       |
+-------------------------------------------------+
Enter choice: 1

=== Recovery Log History ===
+-------+-------------------------------------------------------------+
| No.   | Action Description                                          |
+-------+-------------------------------------------------------------+
| 1     | Viewed Environmental Report                                 |
| 2     | Checked connections from Residential North                  |
| 3     | Viewed waste bin records                                    |
| 4     | System initialized                                          |
+-------+-------------------------------------------------------------+
```

6. Problem Statement / Case Study Results and Conclusion

Analysis of Results

1. **Database Search Speeds:** The AVL tree successfully maintained a balanced height of 4 under successive insertions, whereas the standard BST skewed to a height of

11. This prevents search time degradation, guaranteeing logarithmic efficiency (O(log N)) instead of linear scan delays.
2. **Transportation Savings:** Dijkstra's Algorithm computed an optimal path of 11 km from the Central Depot to the Recycling Center. Compared to standard routes, this reduces simulated fuel consumption and vehicle wear.
3. **Queue fairness:** Utilizing a custom singly-linked FIFO Queue ensures citizen requests are resolved in order, preventing ticket starvation.
4. **Undo Operations:** The Stack recovery log correctly tracks state changes, allowing undo rollbacks in constant O(1) time.

Conclusion

The Smart Waste Collection & Recycling Management System demonstrates how custom pointer structures and algorithms can be integrated to resolve logistical, data storage, and administrative bottlenecks in modern smart cities.
For future extensions, the system can scale by:

IoT Sensor Integration: Feeding live ultrasonic bin readings directly to update the AVL tree nodes.
Live Traffic Weighing: Interfacing map APIs to adjust graph weights dynamically, reflecting congestion.

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7. References

1. GitHub Repos
2. Youtube
3. GeeksforGeeks — *AVL Trees, Graph Traversals, and Hashing Collision Resolutions*.
4. Class Notes